

Kshitij Mishra, Ph.D.

Postdoctoral Associate · Mohamed bin Zayed University of Artificial Intelligence

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[Google Scholar](#) / [GitHub](#) / [LinkedIn](#)

Research profile

I develop adaptive and trustworthy foundation-model agents that learn through interaction, reason strategically, and remain consistent, safe, auditable, and controllable. My work connects reasoning and post-training with conversational AI, reinforcement learning, strategic interaction, and agent safety. My research agenda extends these questions across language and vision-language settings.

Consistent	Evidence-grounded reasoning, factual and behavioral consistency, context attribution.
Adaptive	Teacher-student and multi-agent learning, feedback, continual improvement.
Strategic	Long-horizon reasoning, collaboration, dialogue planning, and persuasion.
Trustworthy	Robustness, prompt-injection defense, safety evaluation, and agent auditing.

Research experience

Postdoctoral Associate Aug 2025 - present

Mohamed bin Zayed University of Artificial Intelligence, Abu Dhabi, UAE

Research on reasoning, strategic interaction, safety alignment, and auditing of language models and tool-using agents.

Developing methods using textual feedback, process-level rewards, and concept-level exploration; building conversational systems and multi-turn evaluations.

Research Intern Feb 2025 - Jun 2025

Quantiphi, Bangalore, India

Investigated reasoning in small language models, with a focus on process-level reward formulations.

Applied Scientist Intern (L5) Jan 2023 - Oct 2023

Amazon.com Services LLC, Los Angeles, USA

Worked on language-model editing and privacy-preserving language models. This work contributed to a Findings of EACL 2024 publication.

Doctoral Researcher Jul 2019 - Aug 2025

Indian Institute of Technology Patna, India

Developed conversational AI for persuasive dialogue, counseling, healthcare, and disability support using reinforcement learning and human feedback.

Contributed to Sevak, an Indian-language chatbot initiative covering railways, judiciary, and healthcare; mentored students and supported teaching in AI, ML, NLP, and theory of computation.

Publications

Published and accepted work · * Equal contribution

- 01 **TRACER: Trace-Regularized Training Against Indirect Prompt Injection.**
Kshitij Mishra, Nils Lukas, and Salem Lahlou.
Findings of EMNLP 2026 · Accepted
- 02 **CORE: Collaborative Reasoning via Cross Teaching.**
Kshitij Mishra, Mirat Aubakirov, Martin Takac, Nils Lukas, and Salem Lahlou.
ICML 2026 · Accepted
- 03 **SD-E²: Semantic Exploration for Reasoning Under Token Budgets.**
Kshitij Mishra, Nils Lukas, and Salem Lahlou.
Findings of EACL 2026
- 04 **Breaking Barriers: A Paradigm Shift in Technology Accessibility for Individuals with Physical Disabilities.**
Kshitij Mishra*, Manisha Burja*, and Asif Ekbal.
AAAI 2025
- 05 **MedEx: Enhancing Medical Question-Answering with First-Order Logic-based Reasoning and Knowledge Injection.**
Aizan Zafar*, **Kshitij Mishra***, and Asif Ekbal.
COLING 2025
- 06 **ABLE: Personalized Disability Support with Politeness and Empathy Integration.**
Kshitij Mishra*, Manisha Burja*, and Asif Ekbal.
EMNLP 2024
- 07 **MedLogic-AQA: Enhancing Medical Question Answering with Abstractive Models Focusing on Logical Structures.**
Aizan Zafar*, **Kshitij Mishra***, and Asif Ekbal.
Findings of EMNLP 2024
- 08 **Correcting Language Model Outputs by Editing Salient Layers.**
Kshitij Mishra*, Tamer Soliman*, Anil Ramakrishna, Aram Galstyan, and Anoop Kumar.
Findings of EACL 2024
- 09 **Please Donate to Save a Life: Inducing Politeness to Handle Resistance in Persuasive Dialogue Agents.**
Kshitij Mishra, Mauajama Firdaus, and Asif Ekbal.
IEEE/ACM Transactions on Audio, Speech, and Language Processing 32 (2024), pp. 2202-2212
- 10 **e-THERAPIST: I suggest you to cultivate a mindset of positivity and nurture uplifting thoughts.**
Kshitij Mishra*, Priyanshu Priya*, Manisha Burja, and Asif Ekbal.
EMNLP 2023
- 11 **PAL to Lend a Helping Hand: Towards Building an Emotion Adaptive Polite and Empathetic Counseling Conversational Agent.**
Kshitij Mishra*, Priyanshu Priya*, and Asif Ekbal.
ACL 2023

Publications

Published and accepted work, continued · * Equal contribution

- 12 **PARTNER: A Persuasive Mental Health and Legal Counselling Dialogue System for Women and Children Crime Victims.**
Priyanshu Priya*, **Kshitij Mishra***, Palak Totala, and Asif Ekbal.
IJCAI 2023
- 13 **RPTCS: A Reinforced Persona-aware Topic-guiding Conversational System.**
Zishan Ahmad*, **Kshitij Mishra***, Asif Ekbal, and Pushpak Bhattacharyya.
EACL 2023
- 14 **Help Me Heal: A Reinforced Polite and Empathetic Mental Health and Legal Counseling Dialogue System for Crime Victims.**
Kshitij Mishra*, Priyanshu Priya*, and Asif Ekbal.
AAAI 2023
- 15 **Genpads: Reinforcing politeness in an end-to-end dialogue system.**
Kshitij Mishra, Mauajama Firdaus, and Asif Ekbal.
PLOS ONE, 2023
- 16 **Predicting Politeness Variations in Goal-Oriented Conversations.**
Kshitij Mishra, Mauajama Firdaus, and Asif Ekbal.
IEEE Transactions on Computational Social Systems 10, no. 3 (2023), pp. 1095-1104
- 17 **PEPDS: A Polite and Empathetic Persuasive Dialogue System for Charity Donation.**
Kshitij Mishra, Azlaan Mustafa Samad, Palak Totala, and Asif Ekbal.
COLING 2022
- 18 **Empathetic Persuasion: Reinforcing Empathy and Persuasiveness in Dialogue Systems.**
Azlaan Mustafa Samad*, **Kshitij Mishra***, Mauajama Firdaus, and Asif Ekbal.
Findings of NAACL 2022, pp. 844-856
- 19 **Please be polite: Towards building a politeness adaptive dialogue system for goal-oriented conversations.**
Kshitij Mishra, Mauajama Firdaus, and Asif Ekbal.
Neurocomputing 494 (2022), pp. 242-254
- 20 **Artificial intelligence for suspended sediment load prediction: a review.**
Deepak Gupta, Barennya Bikash Hazarika, Mohanadhas Berlin, Usha Mary Sharma, and **Kshitij Mishra**.
Environmental Earth Sciences 80, article 346 (2021)
- 21 **Regularized Universum twin support vector machine for classification of EEG Signal.**
Deepak Gupta, Hemanga Jyoti Sarma, **Kshitij Mishra**, and Mukesh Prasad.
IEEE International Conference on Systems, Man and Cybernetics (SMC), 2019, pp. 2298-2304
- 22 **Artificial Neural Networks in Hydrology.**
Kshitij Mishra and Deepak Gupta.
Research Cell: An International Journal of Engineering Science, vol. 31, January 2019, pp. 52-59

Education & academic service

Education

Ph.D., Computer Science and Engineering

Indian Institute of Technology Patna, India

Research: politeness, empathy, personalization, persuasion, and counseling in conversational AI systems.

M.Tech., Computer Science and Engineering

National Institute of Technology Arunachal Pradesh, India · 2017 - 2019

Gold Medalist; CGPA 9.01.

Thesis: Heuristic Learning for an Automated Theorem Prover.

Honors & fellowships

Prime Minister's Research Fellowship (PMRF), Government of India, 2019.

Gold Medal, M.Tech., NIT Arunachal Pradesh, 2019.

GATE Scholarship, Government of India, 2017.

Teaching, mentoring & academic service

Mentored students at multiple academic levels and contributed to teaching and teaching-assistant duties in theory of computation, machine learning, natural language processing, and artificial intelligence at IIT Patna.

Workshop and course organization

2022 · GIAN Workshop on Deep Learning Techniques for Conversational Artificial Intelligence, IIT Patna, April 13-24.

2020 · CEP course on Deep Learning for NLP, IIT Patna, January 10-23.

2018 · Recent Trends in Computing and Computer Networking, NIT Arunachal Pradesh, September 3-7.

2017 · Recent Trends in Machine Learning, Big Data and IoT, NIT Arunachal Pradesh, November 13-17.

Technical skills & languages

Python, C++, C, LaTeX · PyTorch, TensorFlow, scikit-learn, Hugging Face
English and Hindi